

Chapter 2: Transmission and Radio Wave Propagation

Outline



Overview of
Transmission

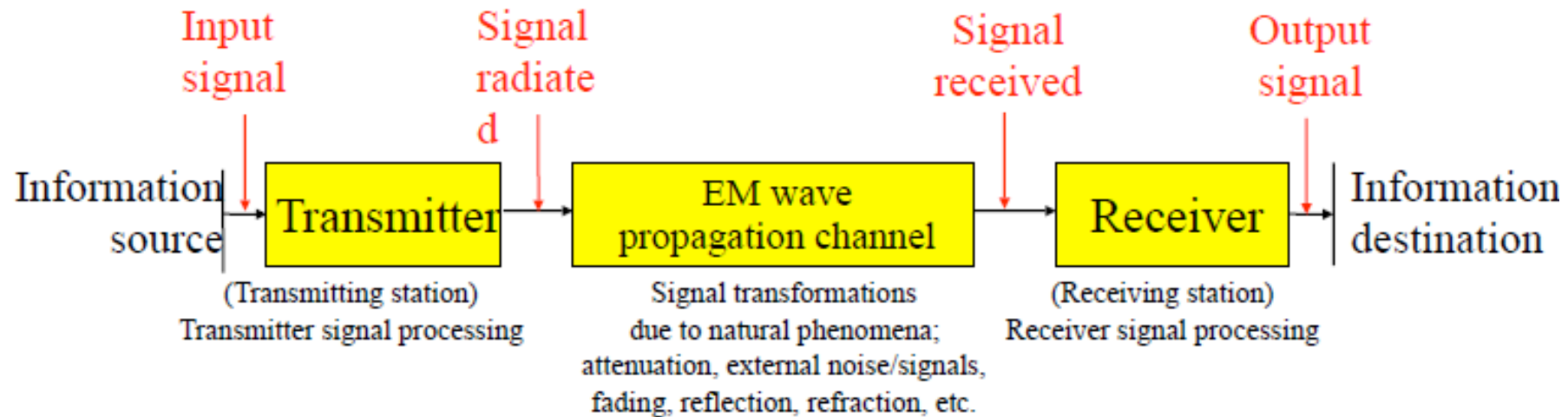
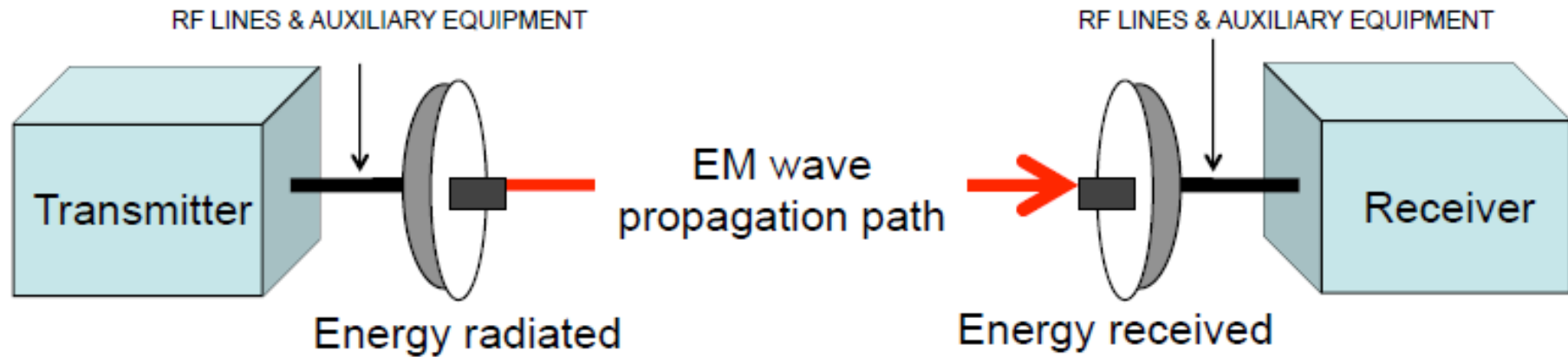


Basic Transmission and
Reception Concepts



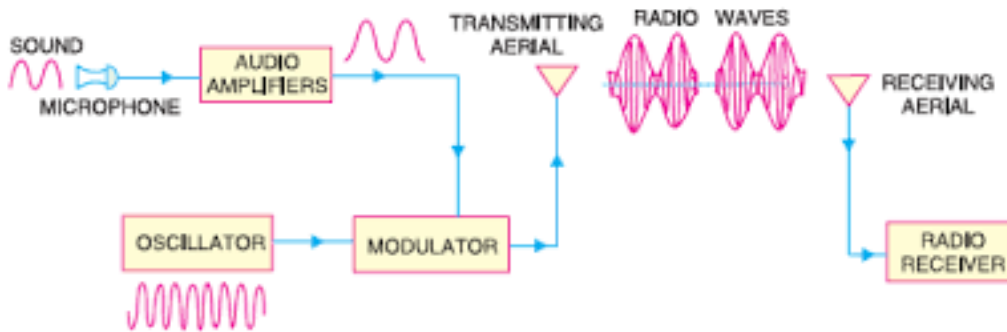
Radio Wave Propagation

Overview of Transmission



Basic Transmission and Reception Concepts

- A **transmitter** is a device that converts information (voice, data, video) into a signal suitable for transmission through a medium (wire, air, or space).



- A **receiver** is a device that captures transmitted signals and converts them back into usable information.

Block	Function
Input Transducer	Converts physical signal (voice/sound) into electrical signal
Signal Processor	Amplifies and processes the signal
Modulator	Imposes information onto a carrier wave
RF Amplifier	Increases signal power for transmission
Antenna	Radiates the signal into space

Block	Function
Antenna	Receives electromagnetic waves
RF Amplifier	Strengthens weak incoming signals
Demodulator	Extracts original information from carrier wave
Signal Processor	Filters and improves signal quality
Output Transducer	Converts electrical signal back to sound/image/data

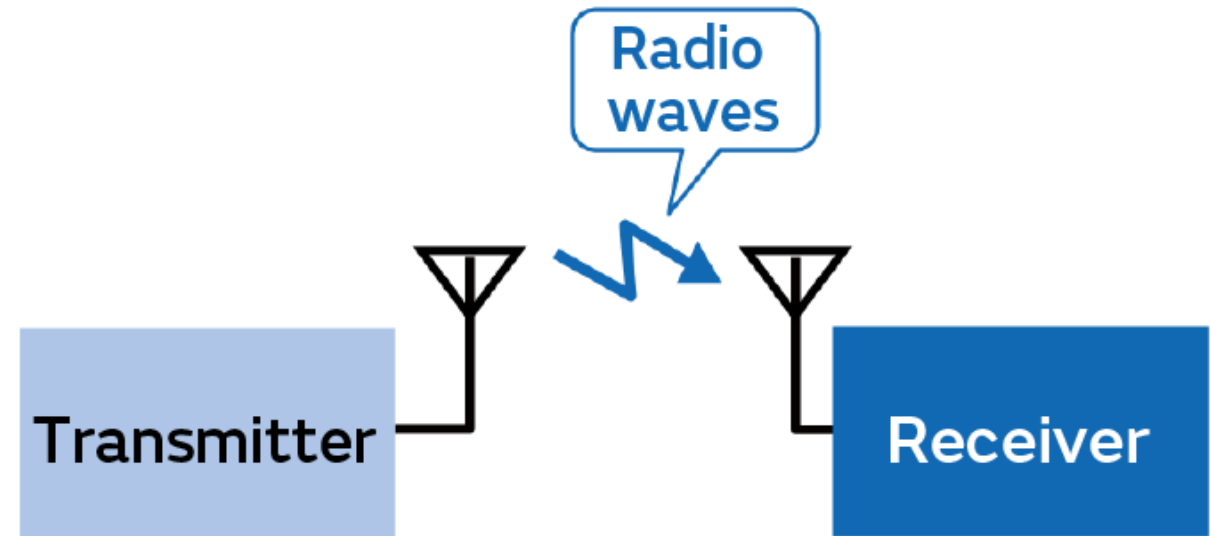
Cont...

Core Conceptual Principles

- The Medium (Channel)
- The Link Budget: The mathematical balance between how much power is sent versus how much is lost.
- The Protocol

Why These Concepts Matter

- Overcome Distance: Using Sky Waves to talk across oceans.
- Increase Capacity: Using high-frequency Space Waves to stream high-definition video.
- Ensure Reliability: Using error-correction to ensure data isn't lost to noise.



Applications of Transmitter & Receiver Concepts

Application Area	How Transmitter & Receiver are Used	Example
 Mobile Communication	Mobile phone transmits voice/data to base station; receiver decodes incoming signals	Calls, SMS, 4G/5G internet
 Radio Broadcasting	High-power transmitter sends audio signals; radios receive and demodulate them	FM/AM radio
 Television Broadcasting	TV station transmits audio + video; TV receiver reconstructs picture and sound	Cable/terrestrial TV
 Satellite Communication	Earth station transmits signals to satellite; satellite relays to receivers on Earth	GPS, DTH TV
 Radar Systems	Transmitter sends pulses; receiver detects reflected signals	Aircraft detection, weather radar
 Wi-Fi Networks	Router transmits data packets; devices receive and decode signals	Internet browsing, streaming
 Navigation Systems	GPS satellites transmit signals; receivers calculate position and time	Google Maps, vehicle tracking

What is Radio Wave Propagation?



Radio wave propagation is the way **radio waves travel through space from a transmitter** (like a radio station or cell tower) **to a receiver** (like your phone or radio).



Key Considerations

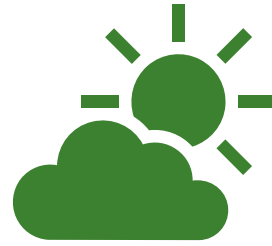
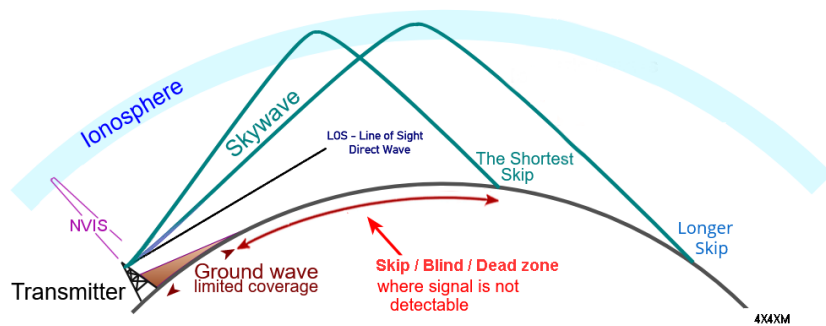
Proper operation: Required signal intensity and Quality of service over required Distance, Area, Volume given the Geographic, climatic region, and time period

Coexistence with other systems - Degradation quality of service and/or service range due to potential radio interference Interference:
Acceptable or not (for both parties)

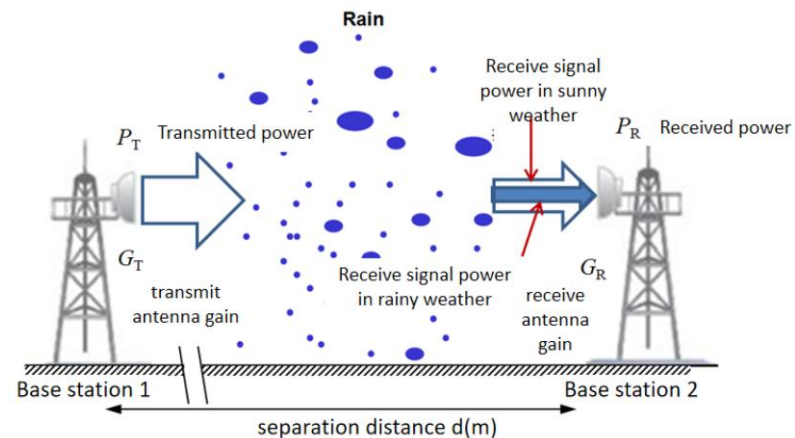
Factors Affecting Propagation



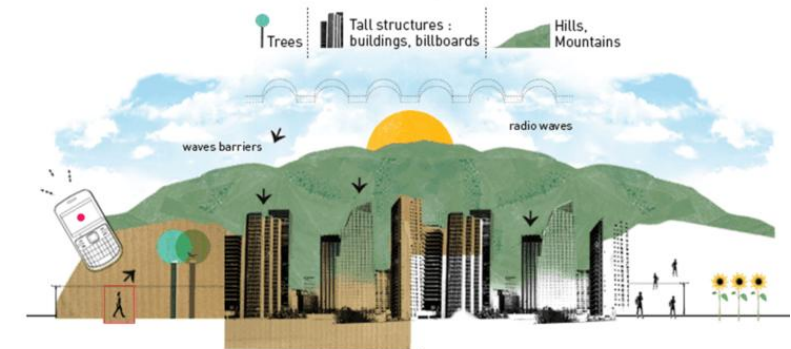
Frequency: Higher frequencies usually travel in straight lines and don't reflect off the ionosphere compared to lower frequencies.



Atmospheric Conditions: Weather, humidity, and solar activity can interfere with signal strength and path.



Obstacles: Buildings, mountains, and trees can block or reflect signals, causing "dead zones."

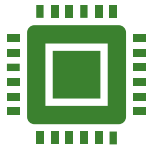


Applications of Propagation Predictions



Wireless Network Planning

Propagation predictions aid in designing efficient wireless networks by optimizing antenna placement and coverage areas.



5G and Beyond

Propagation modeling is crucial for implementing advanced wireless technologies like 5G and future generations to ensure reliable connectivity.



IoT Deployment

Accurate propagation predictions support IoT deployment by enabling optimal sensor placement for seamless connectivity and coverage.



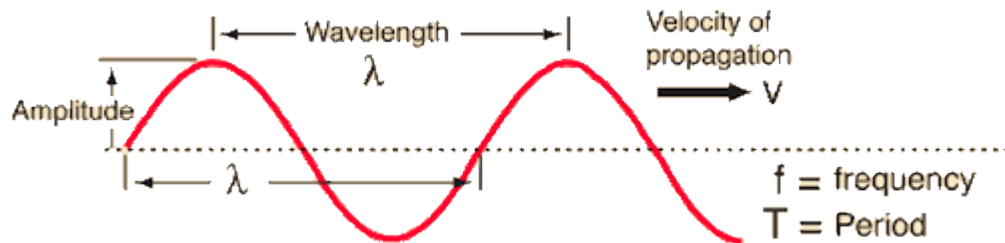
Satellite Communication

Propagation models assist in satellite communication systems by predicting link quality and optimizing ground station locations.

Speed, Wavelength, Frequency

Speed of Light (c) = Wavelength $\times$ Frequency

$$c = 3 \times 10^8 \text{ m/s} = 300,000 \text{ km/s}$$



System	Frequency	Wavelength
AC current	60 Hz	5,000 km
FM radio	100 MHz	3 m
Cellular	800 MHz	37.5 cm
Ka band satellite	20 GHz	15 mm
Ultraviolet light	10^{15} Hz	10^{-7} m

Factors Affecting Radio Propagation



Terrain and Obstacles

Terrain and obstacles such as mountains, buildings, and vegetation can block or reflect radio waves, affecting propagation.



Atmospheric Conditions

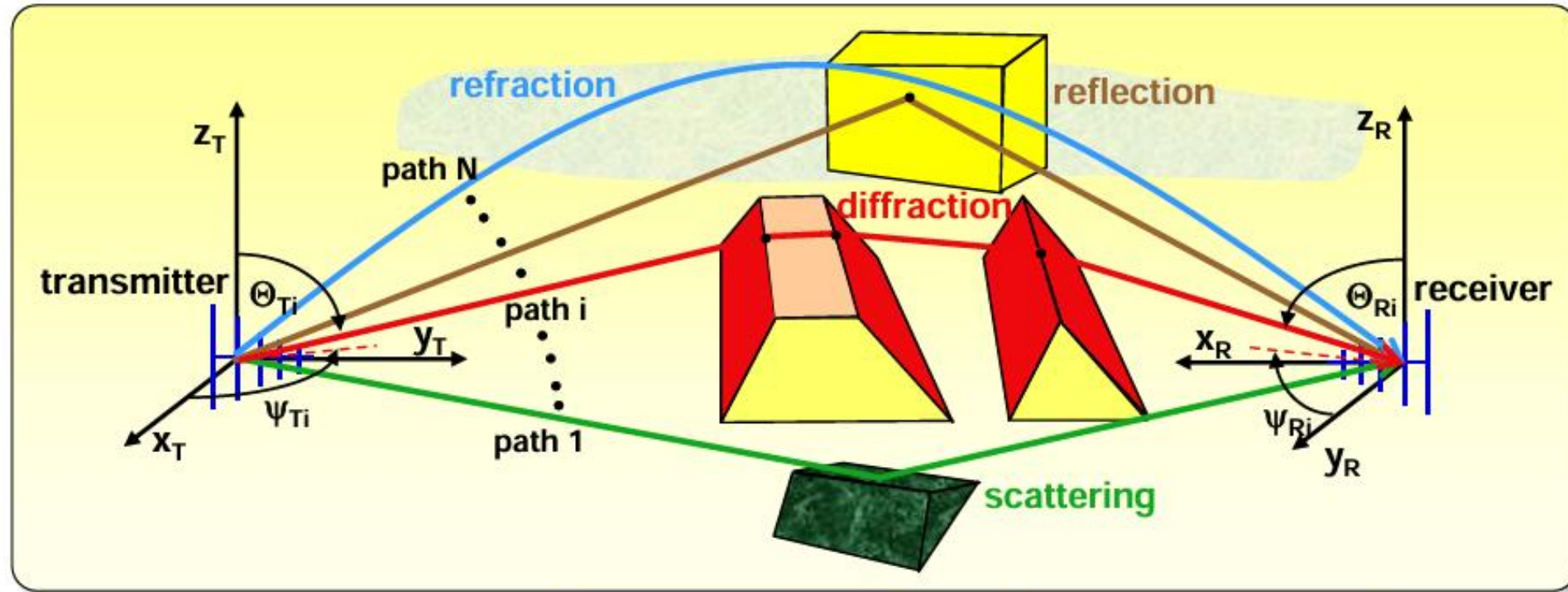
Changes in atmospheric conditions, such as rain, fog, and humidity, can attenuate or scatter radio waves, impacting propagation.



Electromagnetic Interference

Electromagnetic interference from man-made sources or natural phenomena can

Propagation Phenomena



free space propagation:

- line of sight
- no multipath

reflection:

- plane wave reflection
- Fresnel coefficients

diffraction:

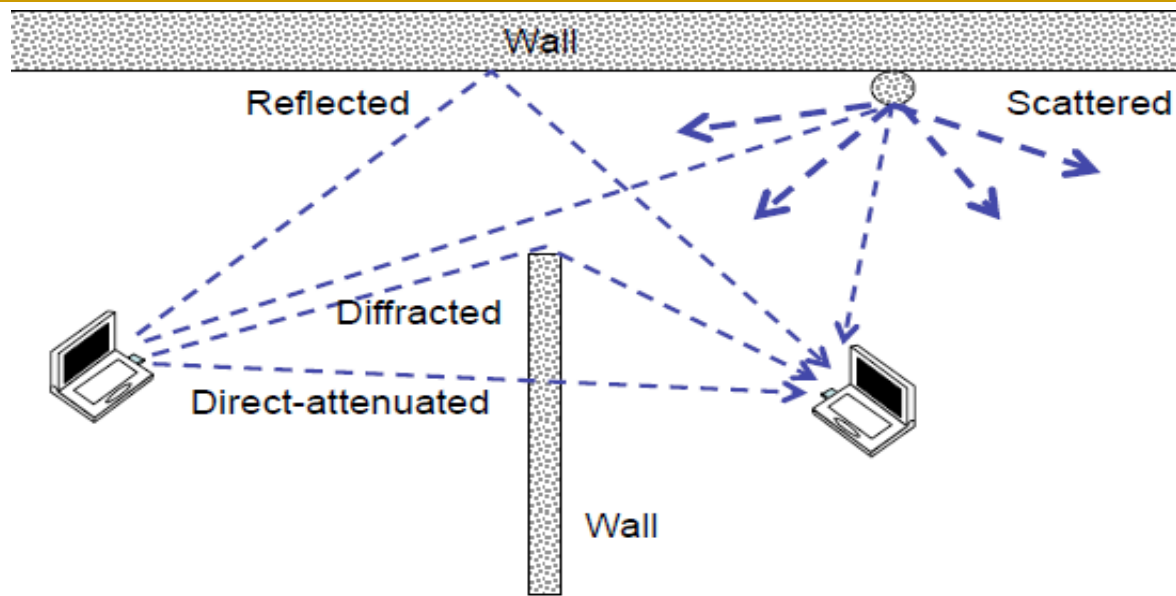
- knife edge diffraction

scattering:

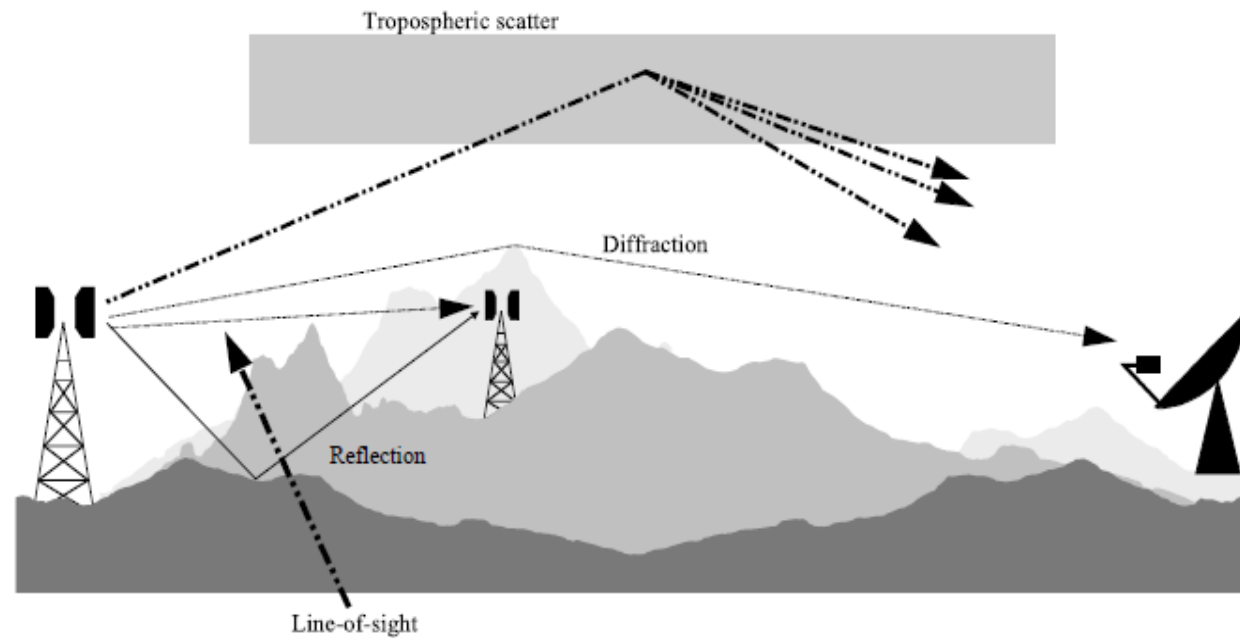
- rough surface scattering
- volume scattering

refraction in the troposphere:

- not considered



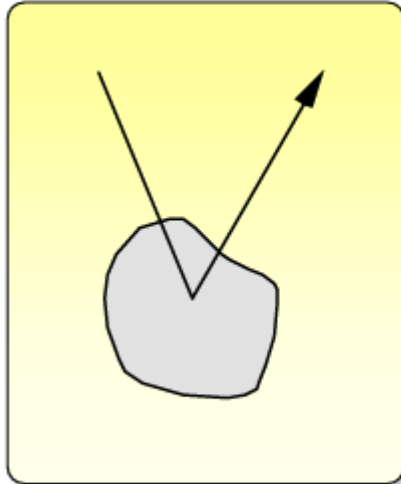
Indoor propagation



Outdoor propagation

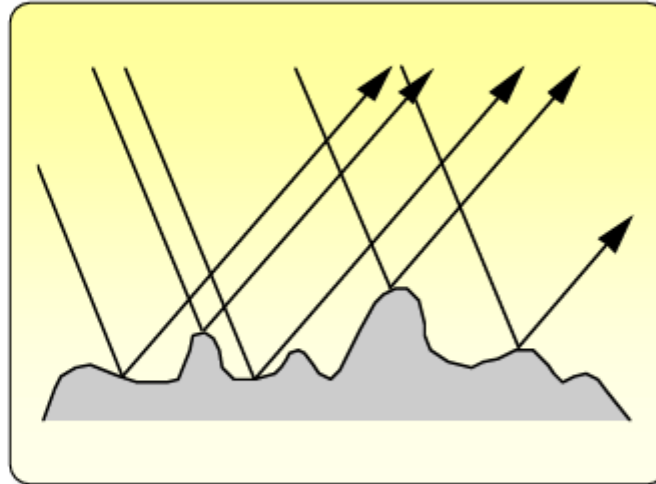
Different Types of Scattering

point scattering

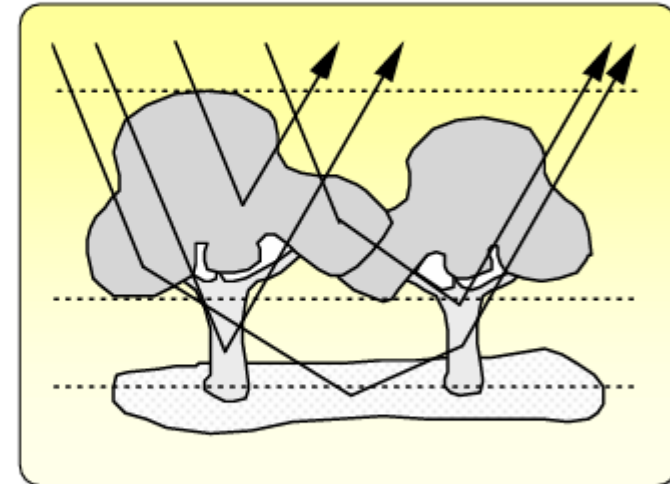


**Simple targets
(plate, sphere,
cylinder, etc.)**

distributed scattering



rough surface scattering

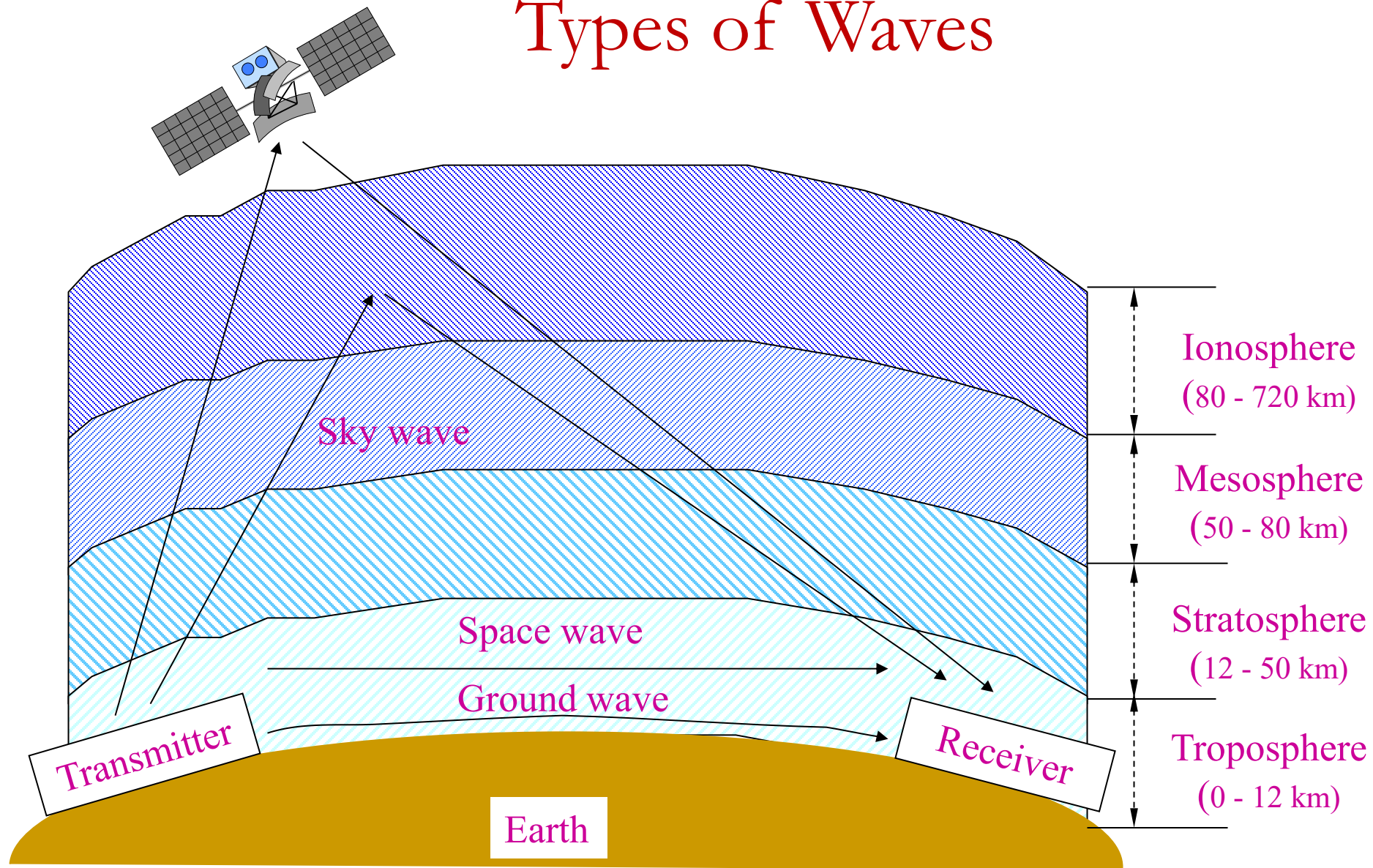


volume scattering

Radio Wave Propagation Table

Band	Frequency Range	Free Space Wavelength Range	Propagation Characteristics	Typical Use
VLF (Very Low Frequency)	3 – 30 kHz	100 – 10 km	Ground wave propagation; follows Earth's curvature; penetrates water well	Submarine communication
LF (Low Frequency)	30 – 300 kHz	10 – 1 km	Strong ground wave; low attenuation; stable signals	Navigation systems, longwave radio
MF (Medium Frequency)	300 kHz – 3 MHz	1 km – 100 m	Ground wave (day), sky wave (night via Ionosphere)	AM radio broadcasting
HF (High Frequency)	3 – 30 MHz	100 – 10 m	Sky wave propagation; reflects off ionosphere for long-distance communication	Shortwave radio, international broadcasting
VHF (Very High Frequency)	30 – 300 MHz	10 – 1 m	Line-of-sight; minimal ionospheric reflection; low noise	FM radio, TV broadcasting
UHF (Ultra High Frequency)	300 MHz – 3 GHz	1 m – 10 cm	Line-of-sight; affected by obstacles; higher data capacity	Mobile phones, Wi-Fi, GPS
SHF (Super High Frequency)	3 – 30 GHz	10 – 1 cm	Highly directional; affected by rain (rain fade); straight-line propagation	Satellite communication, radar
EHF (Extremely High Frequency)	30 – 300 GHz	1 cm – 1 mm	Very high attenuation; limited range; requires clear line-of-sight	Advanced radar, experimental communications

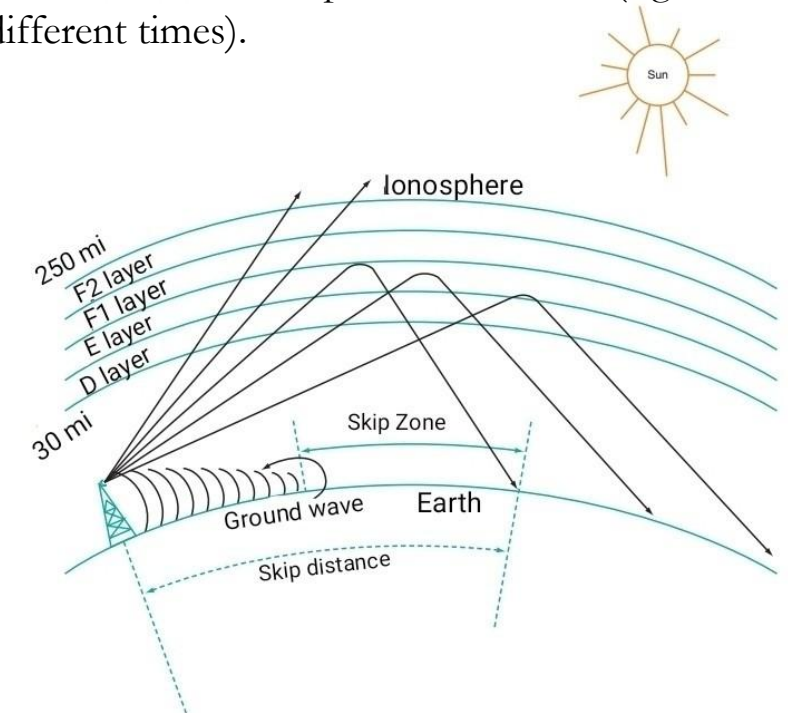
Types of Waves



Sky Wave Propagation

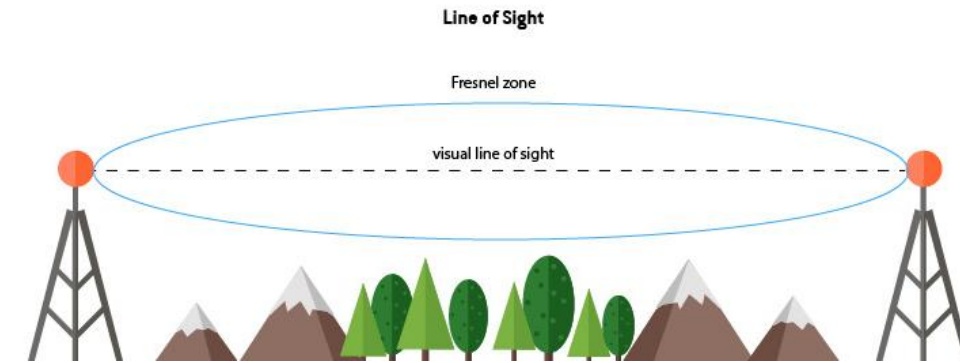
- Sky wave propagation is a mode of radio communication where **electromagnetic waves, directed toward the sky at a specific angle, are reflected or refracted back** to Earth by the ionosphere.
- **Frequency range:**
 - ✓ 3 MHz – 30 MHz (HF band)
 - ✓ Below 3 MHz → waves get absorbed.
 - ✓ Above 30 MHz → waves escape into space
 - ✓ **Ionospheric “reflection”:** Waves don’t actually reflect – they bend (refraction) in the Ionosphere and return to Earth.
 - ✓ **Multi-hop transmission:** Waves can bounce multiple times between Earth and the ionosphere → allows very long-distance communication.
- **Advantages:** Enables massive, global communication ranges without expensive physical infrastructure like satellites or cables.
- **Disadvantages:** Suffers from fading (random fluctuations in signal strength) due to changing atmospheric conditions and multipath interference (signals arriving via different paths at different times).

Layer	Height (km)	Daytime Role	Nighttime Role
D Layer	50 – 90	Absorbs HF signals	Disappears
E Layer	90 – 140	Reflects some MF & HF signals	Weakens or disappears
F Layer	140 – 400	Splits into F1 & F2 layers	Merges into one; main layer for long-distance communication



Space Wave Propagation

- Space wave propagation (also known as **line-of-sight** or tropospheric propagation) occurs when radio waves travel in a **straight line from the transmitting antenna to the receiving antenna**.
- The **radio horizon** is slightly greater than the optical horizon because the atmosphere bends (refracts) radio waves slightly back toward Earth.



Category	Item	Explanation
Applications	FM Radio & TV	Local broadcasting within line-of-sight
	Cellular Networks	Communication between mobile phones and base stations
	Satellite Communication	Signals travel directly to satellites
	Radar	Detects objects using straight-line waves
Limitations	Earth's Curvature	Limits maximum communication distance
	Obstructions	Buildings, mountains, and trees block signals
	Atmospheric Absorption	High frequencies are affected by rain and gases

- **Maximum Line-of-Sight Distance**

$$d = \sqrt{2Rh_t} + \sqrt{2Rh_r}$$

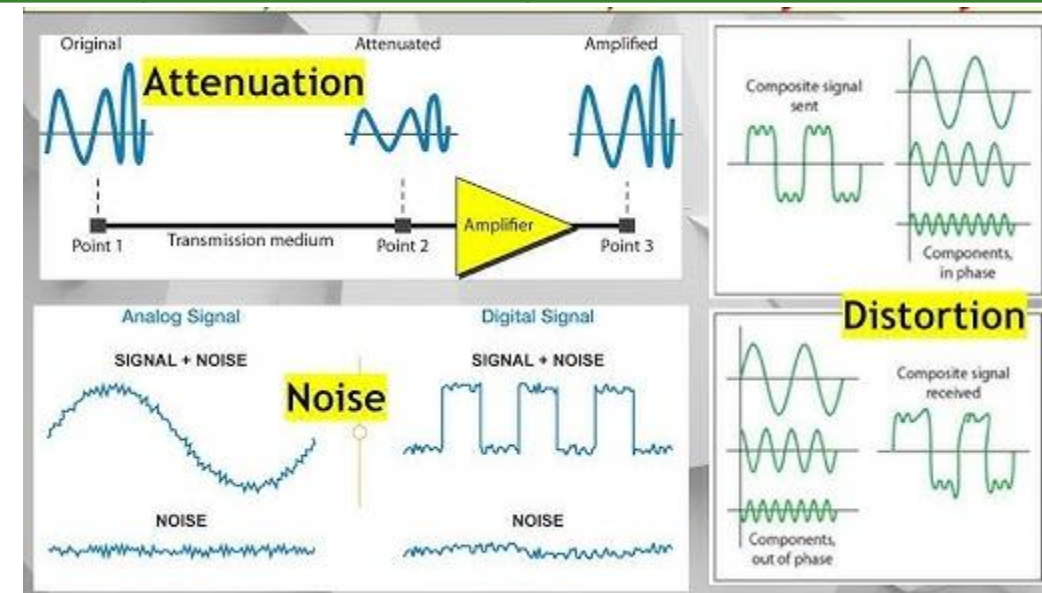
Where:

- d = Maximum line-of-sight distance
- h_t = Height of transmitting antenna
- h_r = Height of receiving antenna
- R = Effective Earth radius

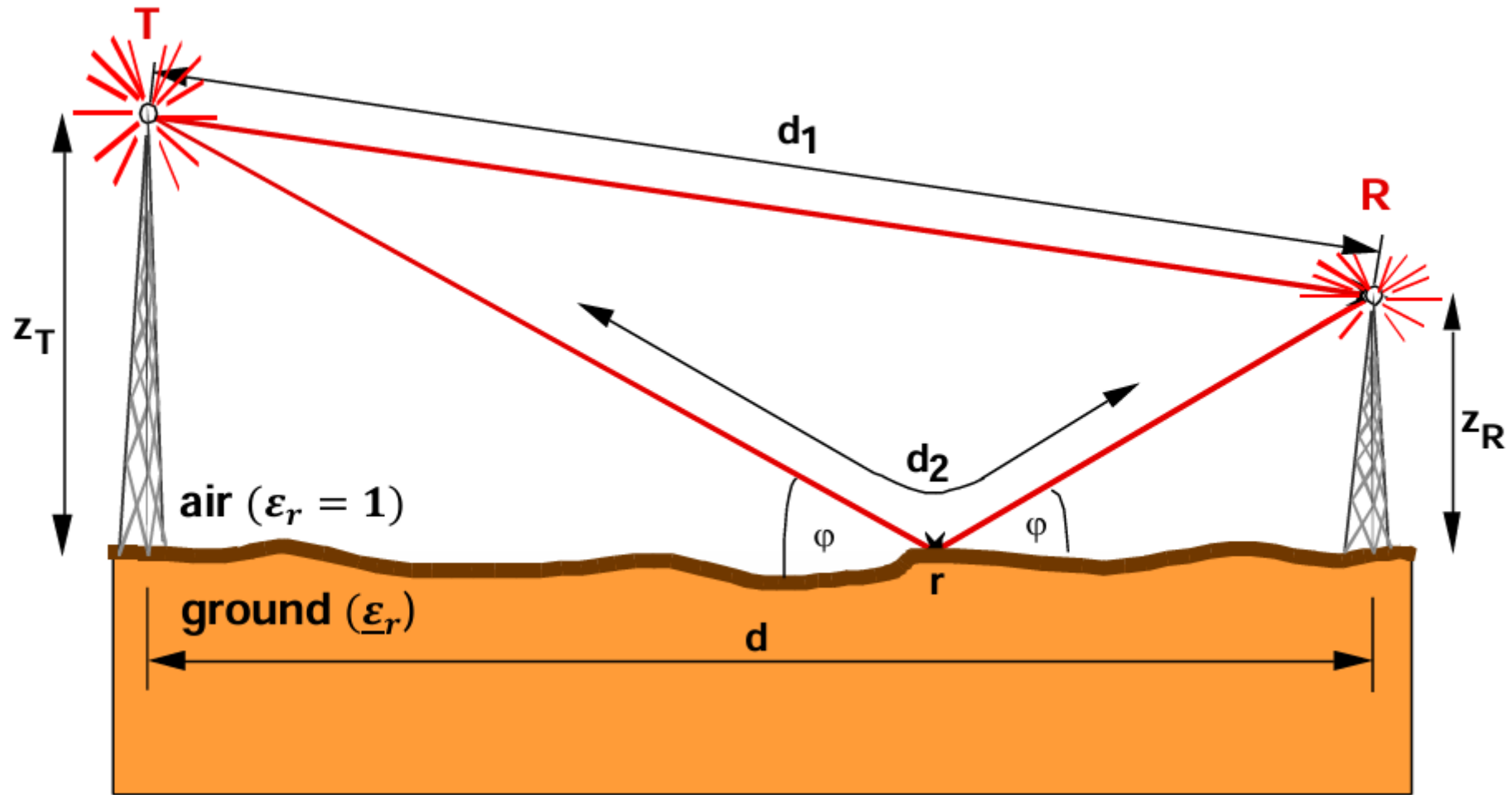
Transmission Line Impairments

Impairment	Meaning	Cause	Effect on Signal	Probable Solutions
Attenuation	Loss of signal strength	Resistance, distance	Signal becomes weak	Amplifiers / Repeaters.
Distortion	Change in signal shape	Different frequencies travel at different speeds	Signal becomes unclear	Use equalizers, proper bandwidth selection, and limit long cable lengths
Noise	Unwanted signals added	External interference, electronics	Errors in transmission	Shielding / Filtering.

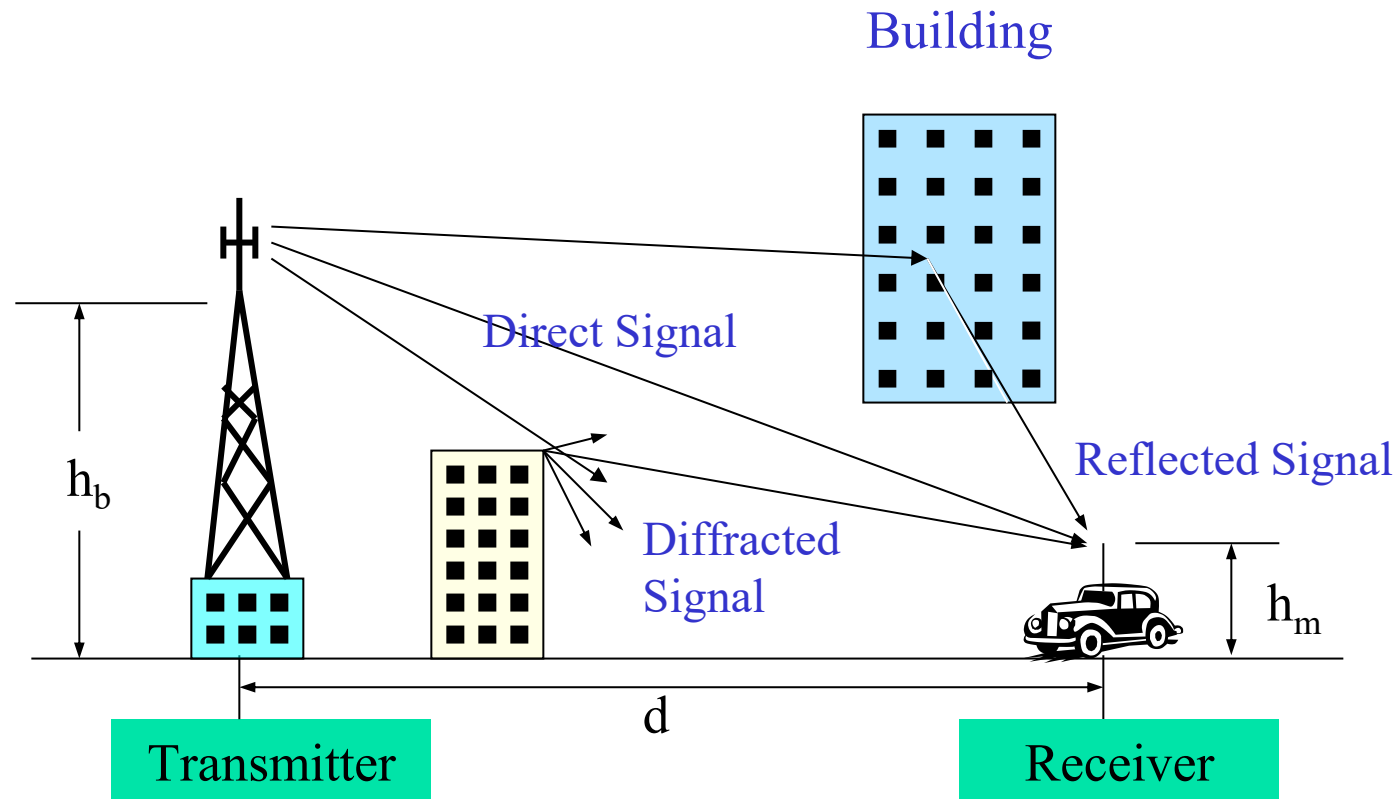
Noise Type	Description
Thermal Noise	Caused by heat in electronic components
Induced Noise	From nearby electrical devices or cables
Crosstalk	Signal interference from nearby wires
Impulse Noise	Sudden spikes (e.g., lightning, switching)



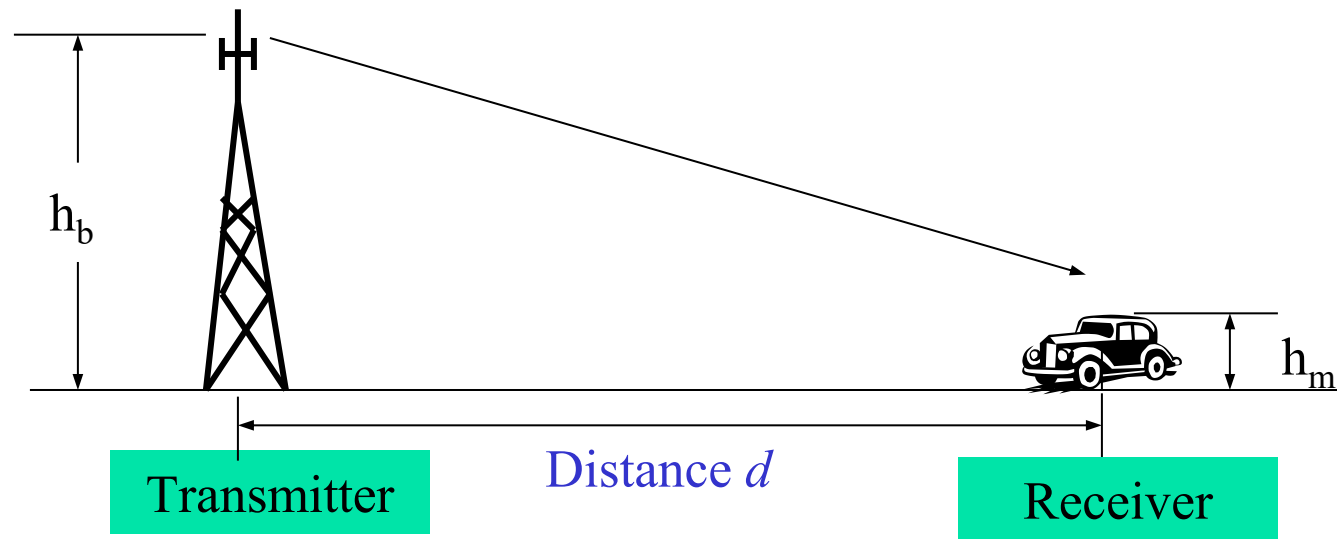
Radio Propagation Effects



Radio Propagation Effects



Example



Free Space Loss

- The **path loss** between a pair of antennas is the **ratio** of the **transmitted power** to the **received power**.
 - ✓ Usually expressed in **dB**.
 - It is basically the **difference** in **dB** between the **transmitted** and **received** signal power:

$$P_L = \frac{P_t}{P_r}$$

$$P_L \text{ dB} = 10 \log_{10} \frac{P_t}{P_r} \text{ dB}$$

- The **dB path gain** is defined as the **negative** of the **dB path loss**;

$$P_G = -P_L = 10 \log_{10} \frac{P_r}{P_t} \text{ dB}$$



The **path loss** depends on:

- Radio frequency
- Distance between transmitter and receiver
- Antenna characteristics (height, location, gains)
- Nature of the terrain (urban/rural, vegetation/clear)
- Propagation medium (dry/moist air)

Friis Transmission Equation

- The power density of a plane wave at distance r is:

$$P_d = \frac{P_t}{4\pi r^2} \text{ (W/m}^2\text{)}$$

- If the transmitting antenna has gain G_t , then:

$$P_d = \frac{P_t G_t}{4\pi r^2}$$

- If the receiving antenna has effective aperture A_e :

$$P_r = P_d \times A_e$$

$$P_r = \frac{P_t G_t}{4\pi r^2} \cdot A_e$$

- Effective aperture of receiving antenna:

$$A_e = \frac{\lambda^2 G_r}{4\pi}$$

Substitute into equation:

$$P_r = \frac{P_t G_t}{4\pi r^2} \cdot \frac{\lambda^2 G_r}{4\pi}$$

- **Friis Transmission Equation (Final Form)**

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi r} \right)^2$$

or using frequency:

$$P_r = P_t G_t G_r \left(\frac{c}{4\pi r f} \right)^2$$

- **Path loss**

$$L = \left(\frac{4\pi r f}{c} \right)^2$$

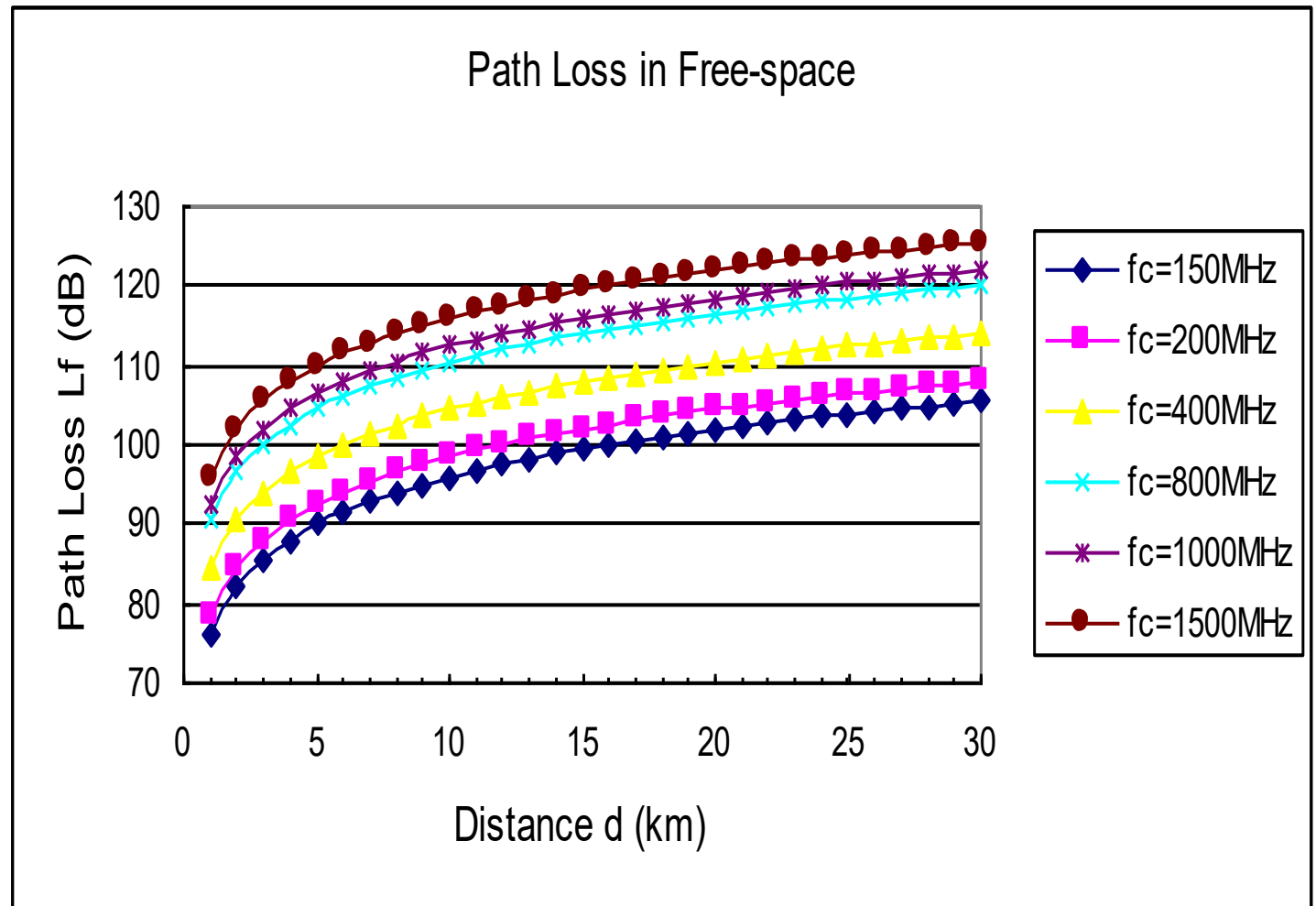
In Decibels (dB form)

$$L_{dB} = 20 \log_{10} F + 20 \log_{10} R - 147.56$$

So,

$$P_r = \frac{P_t G_t G_r}{L}$$

Example of Path Loss (Free-space)



Example

Consider an indoor wireless LAN with $f_c = 900$ MHz, cells of radius 10 m, and nondirectional antennas. Under the free-space path loss model, what transmit power is required at the access point such that all terminals within the cell receive a minimum power of $10 \mu\text{W}$.

How does this change if the system frequency is 5 GHz?

Solution

$$\frac{P_r}{P_t} = \left[\frac{\sqrt{G_l} \lambda}{4\pi d} \right]^2$$

$$P_t = P_r G_l \left[\frac{4\pi d}{\lambda} \right]^2$$

- $G_l = 1$, for non-directional antennas.

$$\lambda = c/f_c = .33 \text{ m}, \quad d = 10 \text{ m}, \quad P_r = 10 \mu\text{W}$$

$$P_t = 10 \mu\text{W} \times 1 \left[\frac{4 \times \pi \times 10 \text{ m}}{0.33 \text{ m}} \right]^2 = 1.45 \text{ W}$$

$$P_t \text{ dB} = 10 \log_{10}[P_t] = 1.61 \text{ dBW dB}$$

- (Recall that P Watts equals $10 \log_{10}[P]$ dBW, dB relative to one Watt, and $10 \log_{10}[P/.001]$ dBm, dB relative to one milliwatt).
- At 5 GHz only,

$$\lambda = .06 \text{ changes,}$$

$$P_t = 43.9 \text{ KW} = 16.42 \text{ dBW.}$$

Example

A transmitter radiates a power of 10 W at a frequency of 1 GHz. The transmitting and receiving antennas both have gains of 2. The separation between the transmitting and receiving antennas is 100 m. Using the Friis transmission equation, determine the received power at the receiving antenna.

Solution

$$P_r = P_t G_t G_r \left(\frac{c}{4\pi r f} \right)^2$$

$$P_r = 10 \times 2 \times 2 \left(\frac{3 \times 10^8}{4\pi \times 100 \times 10^9} \right)^2$$

$$P_r = 40 \times 5.71 \times 10^{-8}$$

$$P_r \approx 2.28 \times 10^{-6} \text{ W}$$

Example

A transmitter radiates 5 W at 2 GHz. Antenna gains are $G_t = 3$, $G_r = 4$, and distance $r = 50m$. Find received power.

Solution

$$P_r = P_t G_t G_r \left(\frac{c}{4\pi r f} \right)^2$$

$$P_r = 5 \times 3 \times 4 \left(\frac{3 \times 10^8}{4\pi \times 50 \times 2 \times 10^9} \right)^2$$

$$P_r = 60 \times 5.71 \times 10^{-8} \approx 3.43 \times 10^{-6} \text{ W}$$

Example

A wireless system operates at two different frequencies, 1 GHz and 2 GHz, while keeping the same transmitter power, antenna gains, and distance in both cases. Using the Friis transmission equation, compare the received power at both frequencies and explain which frequency provides better performance.

Solution

Received power is inversely proportional to the square of the frequency

$$P_r \propto \frac{1}{f^2}$$

- ✓ At **1 GHz** → **higher received power**
- ✓ At **2 GHz** → **lower received power**

1 GHz gives better performance because lower frequency suffers less free-space loss.

Example

In a free-space communication link, the received power is $2 \times 10^{-6} \text{ W}$, the transmitted power is 10 W, the transmitting antenna gain is 2, the receiving antenna gain is 3, and the operating frequency is 1.5 GHz. The distance between the transmitter and receiver is unknown. Using the Friis transmission equation, determine the separation distance.

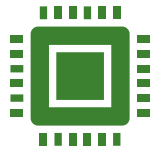
Solution

$$r = \frac{c}{4\pi f} \sqrt{\frac{P_t G_t G_r}{P_r}}$$

$$r = \frac{3 \times 10^8}{4\pi \times 1.5 \times 10^9} \sqrt{\frac{10 \times 2 \times 3}{2 \times 10^{-6}}}$$

$$r = 1.59 \times 10^{-2} \times 5477 \approx 87 \text{ m}$$

Advancements in Propagation Research



Machine Learning-based Models

Machine learning techniques are being applied to develop adaptive propagation models for dynamic and complex environments.



3D Propagation Modeling

Advanced 3D propagation modeling considers elevation data and building structures for more accurate predictions in urban settings.



Millimeter Wave Propagation

Research on millimeter wave propagation aims to harness higher frequencies for enhanced wireless capacity and performance.



Multi-path Mitigation Techniques

Innovative techniques are devised to mitigate multipath effects and improve the accuracy of propagation predictions